

Devarchith Parashara Batchu

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PROFILE

I am a Master's student in Computer Science at UC Davis looking to join your organization as a software or ML engineering intern and contribute through my engineering expertise

EDUCATION

University Of California, Davis

2025 - 2027

Masters in Computer Science

SRM Institute of Science and Technology

2021- 2025

Bachelors in Computer Science Engineering with AI and ML, **GPA: 3.96/4**

SKILLS AND TECHNOLOGIES

Specialized Coursework (Masters): Machine Learning and Discovery, Database Distributed Systems

Specialized Coursework (Bachelors): Artificial Neural Networks, Computer Vision, Artificial Intelligence, Deep Learning, Essentials In Cloud And Devops, Natural Language Processing, Discrete Math, Queuing theory, and Statistics

Programming Languages: Python, C++, JavaScript, TypeScript, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, Debugging

Cloud & DevOps: AWS (EC2, S3), Docker, Linux, Git, CI/CD, PM2

Frameworks & Libraries: React, Next.js, Node.js, Flask, FastAPI, TensorFlow, PyTorch, LangChain, OpenCV

Databases: MySQL, PostgreSQL, MongoDB

WORK EXPERIENCE

QualOrbis Technologies Private Limited

Bengaluru, India

AI/ML Engineer Intern

Dec 2023 - Feb 2024

- Developed Transformer-based NLP microservices, boosting model response accuracy by **14%** and applied DevOps to reduce deployment turnaround from **weeks to days**.
- Refactored ETL pipelines (Python + Pandas) to reduce latency by **30%** and scale processing to **10K** requests/min under peak traffic.
- Deployed BERT-powered sentiment analysis on AWS EC2, enabling emotion-aware dialogue features that increased positive user feedback **8%**.
- Worked with senior engineers to implement CI/CD integrations, enhance monitoring, and support sprint reviews, ensuring **continuous delivery**.

ACADEMIC PROJECTS

Privacy-Preserving Federated Facial Emotion Recognition via Diffusion-Augmented Data and Graph Neural Networks

- Built federated GNN-based architecture for distributed emotion recognition, ensuring privacy across **100+ simulated edge clients**.
- Applied Diffusion Model augmentation, improving decentralized accuracy by **15%** while maintaining **92% mAP**.
- Improved processing speed by **30%** and training time per round by **18%**, improving **total training efficiency 1.5x**.

Wealth Recommendation Conversational Finance Assistant using NLP, Transformer Models, and LangChain Technologies

- Developed an end-to-end finance assistant using React/Next.js with a Node.js API layer and a Flask-based ML inference service for intent understanding.
- Designed and optimized a LangChain-driven RAG pipeline with embedding caching and vector search tuning, reducing retrieval latency by **20%** and improving model insight accuracy by **18%**.
- Scaled an AWS Lambda-based backend (Docker + Node.js) for the multi-tenant finance recommendation system, enabling 2x more concurrent users (~8K) while maintaining a **0.8s average response time**.
- Improved recommendation diversity index by **26%** and end-user satisfaction by **33%**, validated through structured A/B testing.

Safety helmet detection system

- Engineered a real-time YOLOv5x object detection pipeline achieving **88.44% mAP** at **40 FPS** for industrial safety monitoring.
- Optimized the inference pipeline for edge/low-light scenarios to maintain high accuracy in real-world conditions.
- Trained on **5,000+** annotated images with train/test/validation splits to ensure generalization.

Health Care Management System

- Built a full-stack application using React and Node.js for patient registration, appointment scheduling, and basic record management.
- Designed a PostgreSQL database with tables for patients, appointments, doctors, and billing, ensuring clean relational structure and consistent data flow.
- Implemented role-based access control (Admin/Doctor/Patient) with secure sessions, reducing unauthorized access issues and improving workflow clarity across user types.
- Containerized and deployed the application on AWS EC2 using Docker, **increasing uptime, improving scalability, and ensuring stable performance** for multi-user access.

Academic Leadership Roles

Placement Coordinator — SRM Institute of Science & Technology

- Acted as the primary contact for **120+ students**, ensuring fast, accurate communication that reduced placement-related confusion and missed deadlines across the class.
- Improved placement readiness by guiding classmates through resume refinement, interview prep, and role-fit strategy, helping students target positions aligned with their strengths.
- Increased overall preparation efficiency by organizing mock interviews, study groups, aptitude sessions, and curated materials, which boosted students' performance during company evaluations.
- Supported students throughout tests, GDs, and interviews, contributing to a significantly **higher placement rate** and enabling classmates to secure roles that matched their skillsets and career goals.

Model United Nations (MUN) — Delegate

- Led bloc discussions and strategy planning, improving team alignment and helping delegates secure stronger positions during moderated and unmoderated caucuses.
- Recognized for effective **diplomacy**, public speaking, and **crisis-handling**, consistently contributing to productive committee sessions and well-structured debate flow.

PUBLICATIONS AND AWARDS

- Batchu Devarchith Parashara, "Wealth Recommendation Conversational Finance Assistant using NLP, Transformer Models, and LangChain Technologies," *Proceedings of the 5th International Conference on Paradigms of Communication, Computing and Data Analytics (PCCDA 2025)*, Sri Dev Suman Uttarakhand University, Rishikesh, India, January 2025.
- Batchu Devarchith Parashara, "Privacy-Preserving Federated Facial Emotion Recognition via Diffusion-Augmented Data and Graph Neural Networks," *Proceedings of the 4th International Conference on Deep Sciences for Computing and Communications (IconDeepCom-2025)*, SRM Institute of Science and Technology in association with Universiti Putra Malaysia, April 2025. (**Best Paper Award**).
- Batchu Devarchith Parashara, *NeuroTrustICE: A Federated EEG-Guided Framework for Trust-Adaptive and Secure Human-ICE Interfaces* (Currently working on this, going to publish in IEEE Consumer Electronics Magazine).
- Batchu Devarchith Parashara, *HydroGuardNLP: A Weather-Aware and Text-Enriched Framework for Intelligent Dehydration Prediction in Construction Workers* (Currently Working).
- Invited as a **Guest Speaker** at the 1st NEXTGEN AOMSI Symposium, Hyderabad (2024) to deliver a session on Generative AI in Oral & Maxillofacial Surgery.